

Paper-5

AMC (Radiographer)

1. **Ring artefact in CT scan is due to**
 - (a) Failure of detector
 - (b) Patient movement
 - (c) High density material in field
 - (d) Averaging of LAC in a given voxel that is heterogeneous in composition
2. **Ways of hardening X-Ray beams**
 - (a) Filtration
 - (b) Increasing voltage
 - (c) Both
 - (d) None
3. **Right ventricle pumps blood into**
 - (a) Aorta
 - (b) Pulmonary arteries
 - (c) Pulmonary veins
 - (d) SVC
4. **Photoelectric attenuation is**
 - (a) Proportional to cube of atomic number
 - (b) Proportional to square of atomic number
 - (c) Inversely proportional to square of atomic number
 - (d) Inversely proportional to cube of mass number
5. **Electrons interact with matter by**
 - (a) Ionization
 - (b) Excitation
 - (c) Both
 - (d) None
6. **Unit of exposure**
 - (a) Curie
 - (b) Roentgen
 - (c) Rad
 - (d) Gray
7. **Output of X-Ray machine is**
 - (a) Proportional to kV
 - (b) Proportional to square of kV
 - (c) Inversely proportional to square of kV
 - (d) None
8. **Quality factor for X-Ray is**
 - (a) 1
 - (b) 4
 - (c) 5
 - (d) 20
9. **Highly radiosensitive tissue amongst the following**
 - (a) Muscle
 - (b) Bone
 - (c) Epidermis
 - (d) Nerve
10. **Which city topped the Swachh Vayu Sarveksha 2023 Clean Air Survey?**
 - (a) Agra
 - (b) Indore
 - (c) Chandigarh
 - (d) Ahmedabad
11. **National Girl Child Day is celebrated on which day?**
 - (a) 24th January
 - (b) 30th January
 - (c) 8th December
 - (d) 10th February
12. **The currently debated Hoollongapar Gibbon sanctuary is situated in which state?**
 - (a) Tripura
 - (b) Uttar Pradesh
 - (c) Goa
 - (d) Assam
13. **Who has recently inaugurated the first medical college in Nagaland?**
 - (a) Hon'ble CM of Nagaland, Shri Neiphiu Rio
 - (b) Hon'ble deputy CM of Nagaland, Shri T R Zeliang
 - (c) Hon'ble Union Minister of Health & Family Welfare, Dr. Mansukh Mandaviya
 - (d) Hon'ble Health Minister of State, Shri. P. Paiwang Konyak
14. **World Stroke Day is celebrated on which day?**
 - (a) 22 October
 - (b) 24 October
 - (c) 29 October
 - (d) 31 October

15. Who is the present Prime Minister of the State of Palestine?
(a) Mohammad Shtayyeh
(b) Benjamin Netanyahu
(c) Mohmoud Abbas
(d) None of the given
16. "Sarang" is an Indian film festival organized in which country?
(a) Australia (b) Canada
(c) South Korea (d) Germany
17. The United Nations Conference on Trade and Development (UNCTAD) is located at which of the following places?
(a) Geneva (b) Rome
(c) Paris (d) Vienna
18. Which country had won the Cricket World Cup in 2019?
(a) Australia (b) New Zealand
(c) South Africa (d) England
19. Pradhan Mantri Swasthya Suraksha Yojana (PMSSY) Scheme launched in which year?
(a) 2003 (b) 2006
(c) 2008 (d) 2014
20. _____ inaugurated the MedTech Expo-2023 in Gandhinagar.
(a) Dr. Mansukh Mandaviya
(b) Hon'ble CM Bhupendra Patel
(c) Dr. V. K. Paul
(d) None of the given
21. Which type of fire extinguisher is used for petroleum fire as well as electrical equipment fire?
(a) Powder type (b) Liquid type
(c) Foam type (d) A & C both
22. Epoxy resins are used as
(a) Detergents (b) Adhesives
(c) Insecticides (d) Repellents
23. Optic fibres are mainly used for which of the following?
(a) Musical instrument
(b) Weaving
(c) Food industry
(d) Communication
24. National Food Security Act, 2013 (NFSA) provides coverage of _____ rural and _____ urban population.
(a) 25%, 50% (b) 50%, 25%
(c) 75%, 50% (d) 60%, 25%
25. To start the slide show from current slide which short cut key is used?
(a) F5 (b) Alt + F5
(c) Ctrl + F5 (d) Shift + F5
26. RAM stands for
(a) Random Access Memory
(b) Random Active Memory
(c) Random Access Method
(d) Random Mechanism Memory
27. Which shortcut key is used to delete character on the left side of the cursor?
(a) Delete (Del) (b) Backspace
(c) Alt + Delete (d) Ctrl + Delete
28. Which key is used to move the cursor at the end of the line?
(a) Home (b) End
(c) Page down (d) Page Up
29. Firefox is _____
(a) Data Analysis (b) Word Processing
(c) Presentation (d) Internet Browser
30. Which tab of MS Word 2016 is used to change the orientation of the page from portrait to landscape?
(a) Insert (b) Design
(c) Layout (d) View
31. Word count option is available in which tab of MS Word 2016?
(a) References (b) Review
(c) View (d) Mailings
32. If you want to apply formats to selected cell that meet specific criteria based on specific values
(a) Style (b) Format
(c) Condition (d) Conditional Formatting
33. To show present date and time which command is used in MS Excel?
(a) Date() (b) Time()
(c) Now() (d) Date & Time()
34. Principles of radiation protection included all except
(a) Time (b) Distance
(c) Shielding (d) Monitoring

35. Number of thoracic vertebrae
(a) 2 (b) 5
(c) 7 (d) 12
36. Mammography units typically operate at
(a) 25 to 30 kvp (b) 60 to 120 kvp
(c) 50 to 75 kvp (d) 80 to 140 kvp
37. Hounsfield unit zero represents
(a) Water (b) Air
(c) CSF (d) Blood
38. X-Ray tube is made of which glass
(a) Burmese (b) Pyrex
(c) Duralex (d) Gorilla
39. False regarding tungsten
(a) High atomic number making it efficient in beam production
(b) High melting point
(c) Atomic number 94
(d) Melting point 3370 degree Celsius
40. Contrast used in MRI
(a) Iodinated contrast agents
(b) Gadolinium
(c) Barium contrast agents
(d) All
41. RF shielding in MRI magnet room is done using
(a) Bismuth (b) Manganese
(c) Copper (d) Lead
42. Patient position for CT ankle joint
(a) Head first supine
(b) Head first prone
(c) Feet first supine
(d) Feet first prone
43. Photoelectric effect yields
(a) Characteristic radiation
(b) Negative ion
(c) Positive ion
(d) All
44. All increase attenuation except
(a) Energy of radiation
(b) Density of absorber
(c) Atomic number of absorber
(d) Electron per gram of absorber
45. Grid ratio is
(a) Total radiation absorbed by grid
(b) Primary scatter absorbed by grid
(c) Scatter radiation absorbed by grid
(d) Ratio of height of lead strips to distance between them
46. Lateral end of uterine tube opens into
(a) Cervix (b) Uterine cavity
(c) Peritoneal cavity (d) Fornix
47. Shoulder joint is formed by _____ of scapula
(a) Glenoid (b) Coracoid
(c) Acromion (d) Spinous process
48. Barium oral contrast concentration is expressed as
(a) %w/w (b) mOsm/kg
(c) g% (d) mg%
49. Intensity of X-Ray beam that leaves the tube is not uniform because
(a) Saturation voltage
(b) Space charge
(c) Tube shielding
(d) Heel effect
50. Correct as per AERB regulations
(a) Wear TLD below lead apron
(b) Wear TLD at wrist level
(c) TLD provides radiation safety to patient
(d) TLD can be stored in control console room of X-Ray equipment
51. Advantages of USG over CT
(a) No harmful side effects
(b) Relatively cheap
(c) Noninvasive
(d) All of the above
52. A type of social engineering where an attacker sends a fraudulent message designed to trick a person into revealing sensitive information is known as _____.
(a) Phishing attack
(b) Ransomware attack
(c) Denial of Service attack
(d) Man in the middle attack
53. Nonstochastic effect of radiation
(a) Epilation (b) Carcinogenesis
(c) Genetic effect (d) All of the above
54. X Ray photons produced by X-Ray machines are
(a) Heterogenous in energy
(b) Homogenous in energy
(c) Both
(d) None

55. **Bragg peak characteristic is utilized in**
(a) Photon therapy
(b) Neutron therapy
(c) Megavoltage therapy
(d) Electron therapy
56. **Plane dividing body into right and left halves**
(a) Median sagittal (b) Coronal
(c) Axial (d) Mid axillary
57. **Orthopantomogram is taken to rule out lesion in**
(a) Mandible (b) Maxilla
(c) Facial bone (d) Skull
58. **The process by which electrons are emitted from a X-Ray cathode filament**
(a) Ionization
(b) Photoconduction
(c) Thermo luminescence
(d) Thermionic emission
59. **X-Rays are**
(a) Stream of electrons
(b) Stream of positively charged particles
(c) Electromagnetic radiations of high frequency
(d) Stream of uncharged particles
60. **Term neoplasia means**
(a) New Growth (b) New cancer
(c) New lesion (d) New disease
61. **Which of the following needs the longest exposure for an X-Ray image.**
(a) Thoracic (b) Abdomen
(c) Spine (d) Pelvis
62. **X-Ray is recorded on a plate coated with**
(a) Gold halide (b) Silver halide
(c) Copper halide (d) Iron halide
63. **Normal range of creatinine is :**
(a) 0 to 0.4 mg/100 ml
(b) 0.6 to 1.5 mg/100 ml
(c) 2 to 4 mg/ 100 ml
(d) 8 to 15 mg/ 100 ml
64. **Nausea and difficulty in breathing after injecting contrast in a patient for CT scan, what type of shock is he experiencing**
(a) Septic (b) Anaphylactic
(c) Cardiogenic (d) Neurogenic
65. **Function of housing around X-Ray tube**
(a) Keeps heat inside the glass envelope
(b) Promotes electrical shock
(c) Minimizes radiation leakage
(d) Helps with image production
66. **What is the position when a patient lies flat on their back and the head is lower than feet.**
(a) Fowler (b) Trendelenberg
(c) Sims (d) Lateral recumbent
67. **When what instructions to give patient after barium enema?**
(a) Don't drink fluid for 6 hours
(b) With old laxatives
(c) Drink plenty of fluids
(d) Don't monitor your bowel movements
68. **What restricts useful X-Ray beams from causing unnecessary exposure?**
(a) Protective barriers
(b) Collimation
(c) Filtration
(d) Cassette Size
69. **A unit measuring radiation is REM. What does REM stand for?**
(a) Radiation Equivalent Man
(b) Roentgen Early Man
(c) Radioactive early management
(d) Radiation evolved management
70. **Which principle states that ionizing exposure to humans should be as far below the dose limits as practical?**
(a) C/kg (b) OAP
(c) FID (d) ALARA
71. **What is one of the duties of a Radiation Safety Officer?**
(a) Conduct yearly reviews on records on Radiation levels
(b) Conduct quarterly reviews of radiation safety programs
(c) Conduct annual briefings and educational sessions with employees
(d) All of the above
72. **Chemical substances that kill pathogenic bacteria:**
(a) Disinfectant (b) Antiseptics
(c) Chloroform (d) Wet wipes

73. Geometric factor responsible for unequal magnification of different portions of the same object:
 (a) Quantum mottle (b) Noise
 (c) Focal spot (d) Distortion
74. What is the best way to alter quality of an X-Ray beam?
 (a) kVp (b) mAs
 (c) Gray (d) REM
75. Which of the following is not a part of quality assurance program?
 (a) Making sure the equipments are efficient
 (b) Making sure the patient is satisfied
 (c) Consistent quality of radiographic images
 (d) Keeping a low patient dose
76. What does PACS stand for?
 (a) Projection Access Communication Systems
 (b) Printer Analog Computer Systems
 (c) Picture Active Computer System
 (d) Picture Archiving and Communication Systems
77. In the postero anterior chest radiography which of the following should be included on the X-Ray?
 (a) Maximum of 8 posterior ribs
 (b) Exhaled lungs
 (c) Trachea including upper thoracic vertebrae
 (d) Top of iliac crests
78. Stomach protruding through the cardiac orifice through the cleft of diaphragm is known as:
 (a) Small bowel obstruction
 (b) Hiatal hernia
 (c) Pleural effusion
 (d) Appendicitis
79. What does GERD stand for?
 (a) Gall Bladder Esophageal Reflux Dysfunction
 (b) Gastroesophageal Reflux Disease
 (c) Gastroesophageal Reflex Disease
 (d) Gastroesophageal Reflex Dysfunction
80. Emulsion in radiographic film contains:
 (a) Cs halide crystals
 (b) Ca tungsten crystals
 (c) Ag halide crystals
 (d) Ag atoms
81. Filter used in mammography:
 (a) Molybdenum (b) Tungsten
 (c) Copper (d) Aluminium
82. Protective lead apron should have minimum lead equivalence of:
 (a) 0.5 mm lead (b) 1 mm lead
 (c) 2 mm lead (d) 0.25 mm lead
83. TLD badges are made of:
 (a) CaSO_4 (b) LiF
 (c) La, B, O (d) CaF, Mn
84. Maximum field of view which can be obtained with a specific radiographic system is generally limited by the:
 (a) Focal spot size (b) Anode size
 (c) Anode angle (d) Focal length
85. Beam penetration can be increased by increasing:
 (a) kV (b) mAs
 (c) FFD (d) Beam area
86. Low kV are used in some procedures for purpose of:
 (a) Increasing penetration
 (b) Increasing contrast sensitivity
 (c) Decreasing patient exposure
 (d) Decreasing area contrast
87. Changing from 5:1 ratio to 10:1 ratio grid will:
 (a) Decrease patient exposure
 (b) Increase image contrast
 (c) Decrease required kV or mAs
 (d) Decreased X ray tube heating
88. Underprocessing of X-Ray film can result in increased film:
 (a) Sensitivity (b) Contrast
 (c) Fog (d) None
89. Substituting high speed radiographic film for a medium speed results in:
 (a) Higher contrast
 (b) More visibility of detail because of more blurring
 (c) Reduced patient exposure
 (d) Increased quantum noise
90. Factors appropriate for conventional chest X-Ray:
 (a) Low contrast (b) 0.1 mm focal spot
 (c) 120 kV (d) 1:1 ratio grid

91. **Advantages of higher kV are all except:**
 (a) Increased patient exposure
 (b) Reduced X-Ray tube heating
 (c) Shorter exposure times
 (d) Decreased area contrast
92. **Single coated X-Ray film used in all except:**
 (a) Mammography (b) CT scan films
 (c) Skull X Ray (d) Fluoroscopy films
93. **All are true for dental X-Ray films except:**
 (a) Embossed dot on film kept near crown of tooth
 (b) Corners have sharp angles
 (c) Films are protected by lead coils
 (d) Occlusion film size is 2 1/4 x 3"
94. **Efficiency of film washing process is done by estimating:**
 (a) Residual iodine
 (b) Residual silver
 (c) Residual thiosulphate
 (d) Residual chlorine
95. **Annual effective radiation dose for technicians is less than:**
 (a) 30 mSv (b) 100 mSv
 (c) 100 mRem (d) 5 mRem
96. **All are life saving measures in contrast reaction effect:**
 (a) Inj Rantac (b) Inj Adrenaline
 (c) Saline infusion (d) Inj Phenargen
97. **Enteroclysis is an investigation of:**
 (a) Esophagus (b) Small bowel
 (c) Large bowel (d) Hepato biliary tree
98. **CCD means:**
 (a) Charged couple device
 (b) Continuous charged detector
 (c) Conventional charged device
 (d) Continuous conventional device
99. **Target material in an X-Ray tube must have:**
 (a) High atomic number
 (b) High mass number
 (c) High cost
 (d) Low melting point
100. **Interaction of photons with matter include all except:**
 (a) Pair production
 (b) Leakage radiation
 (c) Photoelectric radiation
 (d) Compton scattering

Answer and Explanation

1. **Ans. (a) Failure of detector**
Exp. Ring artefacts in CT scans are usually caused by a failure or miscalibration of one or more detectors.
2. **Ans. (c) Both**
Exp. Ways to harden X-ray beams include using filtration and increasing the voltage.
3. **Ans. (b) Pulmonary arteries**
Exp. The right ventricle pumps blood into the pulmonary arteries.
4. **Ans. (a) Proportional to cube of atomic number**
Exp. Photoelectric attenuation is proportional to the cube of the atomic number (Z^3).
5. **Ans. (c) Both**
Exp. Electrons interact with matter by ionization and excitation.
6. **Ans. (b) Roentgen**
Exp. The unit of exposure is the Roentgen (R).
7. **Ans. (b) Proportional to square of kV**
Exp. The output of an X-ray machine is proportional to the square of the kilovoltage (kV^2).
8. **Ans. (a) 1**
Exp. The quality factor for X-rays is 1.
9. **Ans. (c) Epidermis**
Exp. The epidermis is highly radiosensitive compared to muscle, bone, and nerve tissue.
10. **Ans. (b) Indore**
Exp. Indore topped the Swachh Vayu Sarvekshan 2023 Clean Air Survey.
11. **Ans. (a) 24th January**
Exp. National Girl Child Day is celebrated on 24th January.
12. **Ans. (d) Assam**
Exp. The Hoollongapar Gibbon Sanctuary is situated in Assam.
13. **Ans. (a) Hon'ble CM of Nagaland, Shri Neiphiu Rio**
Exp. Shri Neiphiu Rio, Hon'ble CM of Nagaland, recently inaugurated the first medical college in Nagaland.
14. **Ans. (c) 29 October**
Exp. World Stroke Day is celebrated on 29 October.
15. **Ans. (a) Mohammad Shtayyeh**
Exp. Mohammad Shtayyeh is the present Prime Minister of the State of Palestine.

16. **Ans. (c) South Korea**
Exp. "Sarang" is an Indian film festival organized in South Korea.
17. **Ans. (a) Geneva**
Exp. The United Nations Conference on Trade and Development (UNCTA (d) is located in Geneva.
18. **Ans. (d) England**
Exp. England won the Cricket World Cup in 2019.
19. **Ans. (a) 2003**
Exp. The Pradhan Mantri Swasthya Suraksha Yojana (PMSSY) Scheme was launched in 2003.
20. **Ans. (a) Dr. Mansukh Mandaviya**
Exp. Dr. Mansukh Mandaviya inaugurated the MedTech Expo-2023 in Gandhinagar.
21. **Ans. (a) Powder type**
Exp. Powder type fire extinguishers are used for petroleum fires as well as electrical equipment fires.
22. **Ans. (b) Adhesives**
Exp. Epoxy resins are used as adhesives.
23. **Ans. (d) Communication**
Exp. Optic fibres are mainly used for communication.
24. **Ans. (c) 75%, 50%**
Exp. The National Food Security Act, 2013 (NFSA) provides coverage of 75% of the rural and 50% of the urban population.
25. **Ans. (d) Shift + F5**
Exp. To start the slide show from the current slide, the shortcut key Shift + F5 is used.
26. **Ans. (a) Random Access Memory**
Exp. RAM stands for Random Access Memory.
27. **Ans. (b) Backspace**
Exp. The backspace key is used to delete a character on the left side of the cursor.
28. **Ans. (b) End**
Exp. The End key is used to move the cursor to the end of the line.
29. **Ans. (d) Internet Browser**
Exp. Firefox is an Internet Browser.
30. **Ans. (c) Layout**
Exp. The Layout tab of MS Word 2016 is used to change the orientation of the page from portrait to landscape.
31. **Ans. (b) Review**
Exp. The word count option is available in the Review tab of MS Word 2016.
32. **Ans. (d) Conditional Formatting**
Exp. Conditional Formatting is used to apply formats to selected cells that meet specific criteria based on specific values.
33. **Ans. (c) Now()**
Exp. The Now() command is used in MS Excel to show the present date and time.
34. **Ans. (d) Monitoring**
Exp. The principles of radiation protection include time, distance, and shielding, but not monitoring, which is a means of measuring exposure, not a protective principle.
35. **Ans. (d) 12**
Exp. There are 12 thoracic vertebrae.
36. **Ans. (a) 25 to 30 kvp**
Exp. Mammography units typically operate at 25 to 30 kvp.
37. **Ans. (a) Water**
Exp. Hounsfield unit zero represents water.
38. **Ans. (b) Pyrex**
Exp. An X-ray tube is made of Pyrex glass.
39. **Ans. (c) Atomic number 94**
Exp. The atomic number of tungsten is not 94; it is 74.
40. **Ans. (b) Gadolinium**
Exp. Gadolinium is used as a contrast agent in MRI.
41. **Ans. (c) Copper**
Exp. RF shielding in an MRI magnet room is done using copper.
42. **Ans. (c) Feet first supine**
Exp. The patient position for CT of the ankle joint is feet first supine.
43. **Ans. (d) All**
Exp. Photoelectric effect results in the emission of characteristic radiation, formation of positive ions, and also involves the ejection of photoelectrons.
44. **Ans. (a) Energy of radiation**
Exp. Increasing the energy of the radiation reduces attenuation, as higher energy X-rays can penetrate materials more effectively.
45. **Ans. (d) Ratio of height of lead strips to distance between them**
Exp. Grid ratio is defined as the ratio of the height of the lead strips to the distance between them.
46. **Ans. (c) Peritoneal cavity**
Exp. The lateral end of the uterine tube (fallopian tube) opens into the peritoneal cavity.

47. **Ans. (a) Glenoid**

Exp. The shoulder joint is formed by the glenoid cavity of the scapula and the head of the humerus.

48. **Ans. (a) %w/w**

Exp. The concentration of barium oral contrast is usually expressed as a percentage weight/weight (%w/w).

49. **Ans. (d) Heel effect**

Exp. The intensity of the X-ray beam that leaves the tube is not uniform because of the heel effect.

50. **Ans. (a) Wear TLD below lead apron**

Exp. According to AERB regulations, TLD should be worn at chest level under the lead apron to accurately measure the dose received by the radiologist.

51. **Ans. (d) All of the above**

Exp. Ultrasound (USG) has no harmful side effects, is relatively cheap, and is noninvasive.

52. **Ans. (a) Phishing attack**

Exp. A phishing attack is a type of social engineering where an attacker sends a fraudulent message designed to trick a person into revealing sensitive information.

53. **Ans. (a) Epilation**

Exp. Nonstochastic (deterministic) effects of radiation include effects like epilation (hair loss), whereas stochastic effects include carcinogenesis and genetic effects.

54. **Ans. (a) Heterogenous in energy**

Exp. X-ray photons produced by X-ray machines are heterogeneous in energy, meaning they have a range of energies.

55. **Ans. (a) Photon therapy**

Exp. The Bragg peak characteristic is utilized in photon therapy, where the energy deposition increases to a peak at a certain depth in tissue.

56. **Ans. (a) Median sagittal**

Exp. The plane dividing the body into right and left halves is the median sagittal plane.

57. **Ans. (a) Mandible**

Exp. An orthopantomogram (OPG) is taken to rule out lesions in the mandible.

58. **Ans. (d) Thermionic emission**

Exp. The process by which electrons are emitted from an X-ray cathode filament is thermionic emission.

59. **Ans. (c) Electromagnetic radiations of high frequency**

Exp. X-rays are electromagnetic radiations of high frequency.

60. **Ans. (a) New Growth**

Exp. The term neoplasia means new growth.

61. **Ans. (b) Abdomen**

Exp. The abdomen generally requires the longest exposure time for an X-ray image due to its density and size.

62. **Ans. (b) Silver halide**

Exp. X-ray films are coated with silver halide.

63. **Ans. (b) 0.6 to 1.5 mg/100 ml**

Exp. The normal range of creatinine is 0.6 to 1.5 mg/100 ml.

64. **Ans. (b) Anaphylactic**

Exp. Nausea and difficulty in breathing after injecting contrast indicate an anaphylactic shock.

65. **Ans. (c) Minimizes radiation leakage**

Exp. The function of the housing around the X-ray tube is to minimize radiation leakage.

66. **Ans. (b) Trendelenberg**

Exp. The position where a patient lies flat on their back with the head lower than the feet is called the Trendelenberg position.

67. **Ans. (c) Drink plenty of fluids**

Exp. After a barium enema, patients are instructed to drink plenty of fluids to help pass the barium.

68. **Ans. (b) Collimation**

Exp. Collimation restricts useful X-ray beams to the area of interest, reducing unnecessary exposure.

69. **Ans. (a) Radiation Equivalent Man**

Exp. REM stands for Radiation Equivalent Man, a unit of measuring radiation.

70. **Ans. (d) ALARA**

Exp. The ALARA principle (As Low As Reasonably Achievable) states that ionizing radiation exposure should be kept as far below dose limits as practical.

71. **Ans. (d) All of the above**

Exp. One of the duties of a Radiation Safety Officer includes conducting yearly reviews of records on radiation levels, quarterly reviews of radiation safety programs, and annual briefings and educational sessions with employees.

72. **Ans. (a) Disinfectant**

Exp. Chemical substances that kill pathogenic bacteria are called disinfectants.

73. **Ans. (d) Distortion**

- Exp.** The geometric factor responsible for unequal magnification of different portions of the same object is distortion.
- 74. Ans. (a) kVp**
- Exp.** The best way to alter the quality of an X-ray beam is by adjusting the kilovoltage peak (kVp).
- 75. Ans. (b) Making sure the patient is satisfied**
- Exp.** Ensuring patient satisfaction is not typically a part of a quality assurance program, which focuses on equipment efficiency, consistent image quality, and keeping patient dose low.
- 76. Ans. (d) Picture Archiving and Communication Systems**
- Exp.** PACS stands for Picture Archiving and Communication Systems.
- 77. Ans. (c) Trachea including upper thoracic vertebrae**
- Exp.** In posteroanterior chest radiography, the trachea including the upper thoracic vertebrae should be included in the X-ray.
- 78. Ans. (b) Hiatal hernia**
- Exp.** Stomach protruding through the cardiac orifice through the cleft of the diaphragm is known as a hiatal hernia.
- 79. Ans. (b) Gastroesophageal Reflux Disease**
- Exp.** GERD stands for Gastroesophageal Reflux Disease.
- 80. Ans. (c) Ag halide crystals**
- Exp.** Emulsion in radiographic film contains Ag (silver) halide crystals.
- 81. Ans. (a) Molybdenum**
- Exp.** The filter used in mammography is typically made of molybdenum.
- 82. Ans. (d) 0.25 mm lead**
- Exp.** A protective lead apron should have a minimum lead equivalence of 0.25 mm lead.
- 83. Ans. (b) LiF**
- Exp.** TLD (Thermoluminescent Dosimeter) badges are commonly made of lithium fluoride (LiF).
- 84. Ans. (c) Anode angle**
- Exp.** The maximum field of view that can be obtained with a specific radiographic system is generally limited by the anode angle.
- 85. Ans. (a) kV**
- Exp.** Beam penetration can be increased by increasing the kilovoltage (kV).
- 86. Ans. (b) Increasing contrast sensitivity**
- Exp.** Low kV is used in some procedures to increase contrast sensitivity.
- 87. Ans. (b) Increase image contrast**
- Exp.** Changing from a 5:1 ratio to a 10:1 ratio grid will increase image contrast.
- 88. Ans. (c) Fog**
- Exp.** Underprocessing of X-ray film can result in increased film fog.
- 89. Ans. (c) Reduced patient exposure**
- Exp.** Substituting high-speed radiographic film for a medium-speed film results in reduced patient exposure.
- 90. Ans. (c) 120 kV**
- Exp.** Factors appropriate for conventional chest X-ray include the use of 120 kV.
- 91. Ans. (a) Increased patient exposure**
- Exp.** Using higher kV has many advantages, but it does increase patient exposure.
- 92. Ans. (c) Skull X-ray**
- Exp.** Single-coated X-ray film is used in mammography, CT scan films, and fluoroscopy films but not typically in skull X-rays.
- 93. Ans. (b) Corners have sharp angles**
- Exp.** Dental X-ray films have rounded corners, not sharp angles.
- 94. Ans. (c) Residual thiosulphate**
- Exp.** The efficiency of the film washing process is determined by estimating the residual thiosulphate.
- 95. Ans. (a) 30mSv**
- 96. Ans. (a) Inj Rantac**
- Exp.** Inj Rantac is not a life-saving measure in contrast reaction effect; Adrenaline, saline infusion, and Inj Phenargen are.
- 97. Ans. (b) Small bowel**
- Exp.** Enteroclysis is an investigation of the small bowel.
- 98. Ans. (a) Charged couple device**
- Exp.** CCD stands for Charged Couple Device.
- 99. Ans. (a) High atomic number**
- Exp.** The target material in an X-ray tube must have a high atomic number.
- 100. Ans. (b) Leakage radiation**
- Exp.** Leakage radiation is not an interaction of photons with matter; interactions include pair production, photoelectric effect, and Compton scattering.